CUSTOMER CAPABILITY STATEMENT

瘤胃胶囊与螺旋蝇蛆病监测能力答复

Rumen Bolus Monitoring Capability for Screwworm (New World Screwworm) — Customer Reply

河南快瑞科技股份有限公司

致：墨西哥牧业客户（To: Our Partner in Mexico）

自：HKT 智慧畜牧团队（From: HKT Smart Livestock Team）

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开篇信 / Opening Note

尊敬的合作伙伴：

感谢您们分享墨西哥牧业当前面临的严峻局势。我们了解到，由于一种苍蝇的幼虫（您们所称的“穿孔虫”，学名新大陆螺旋蝇 Cochliomyia hominivorax）在牛只伤口中寄生，美国已关闭对墨西哥活牛的进口边境，这给您们的出口业务带来了直接冲击。这封信用尽可能简单的语言，如实说明我们的瘤胃胶囊监测系统在这个问题上能做到什么、不能做到什么——我们相信只有先讲清楚边界，合作才能长久。

Thank you for sharing the serious situation facing the Mexican cattle industry. We understand that the US border closure to Mexican feeder cattle, driven by New World Screwworm (NWS, Cochliomyia hominivorax), has hit your export business directly. This letter states plainly what our rumen bolus monitoring system can and cannot do — we believe a durable partnership starts with honest boundaries.

一句话总结：胶囊不能看到伤口上的蛆虫，但它能每天 24 小时盯着每一头牛，在牛因感染而发热、吃不下饭、喝水变得不规律的时候发出提醒，让您的牛仔在人眼还没注意到之前就把那头牛找出来、尽早治疗。 In one sentence: the bolus cannot see maggots on a wound, but it watches every animal 24/7 and alerts you when fever, loss of appetite or changed drinking patterns signal infection — so your cowboys find and treat that animal earlier than a once-a-day visual check would.

—— HKT 智慧畜牧团队 / HKT Smart Livestock Team

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一、结论速览 1. Summary at a Glance

我们的瘤胃胶囊是一枚牛吞下后存留在瘤胃里的传感器胶囊，连续测量每头牛的体温和活动，每 30 分钟自动上报一次，电池可用约 5 年。它是一位不知疲倦的连续监护员，而不是医生。 Our rumen bolus is a sensor capsule that stays in the reticulorumen and continuously measures each animal's temperature and activity, reporting every 30 minutes on a roughly 5-year battery. Think of it as a tireless continuous sentinel, not a veterinarian.

表 1 能力速览总表 / Table 1 Capability summary

问题 / Question	结论 / Verdict
胶囊能直接看到伤口和蛆虫吗？ / Can the bolus directly see wounds or maggots?	不能。胶囊在瘤胃里，看不到体表。 / No. It sits inside the rumen and cannot see the skin surface.
能提醒“这头牛生病了，快去看它”吗？ / Can it flag “this animal is sick, go check it”?	可以。发热、食欲下降、饮水异常、活动减少等综合信号会触发告警。 / Yes. Fever, appetite loss, abnormal drinking and reduced activity trigger alerts.
能确诊“这就是蛆病”吗？ / Can it diagnose screwworm specifically?	不能。发热可能来自任何疾病，确诊需人工检查伤口。 / No. Fever can come from any disease; confirm by wound inspection.
能替代美墨两国的官方检疫吗？ / Can it replace official border inspection?	不能。检疫以官方规则为准；监测记录只能作为牛只健康档案的辅助材料。 / No. Official protocols govern; monitoring records are only supporting health records.
犊牛（最常见的感染群体之一）可以用胶囊吗？ / Can calves — one of the most commonly affected groups — use the bolus?	不适用。幼牛瘤胃尚未发育完全，建议用我们的耳标监测仪。 / Not suitable. Calves need our ear-tag device instead.

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二、我们能做到的：让病牛更早被找到 2. What We Deliver: Finding Sick Animals Earlier

蛆病发展分两个阶段。早期（苍蝇刚产完卵，伤口小）牛只看上去一切正常，任何仪器都无法察觉；到了中期，幼虫啃食组织引起炎症和继发感染，牛开始发热、

吃不下饭、精神沉郁、饮水不规律——这些身体反应正好落在胶囊能测的范围里。
Screwworm develops in two stages. Early on, right after egg hatch, the animal looks completely normal and no instrument can detect it. By mid-stage, larvae feeding on tissue cause inflammation and secondary infection; the animal develops fever, loses appetite, becomes depressed and drinks irregularly. These whole-body reactions are exactly what the bolus measures.

2.1 四类告警，四个早期线索 2.1 Four alert types, four early clues

表 2 病牛身体反应与告警对照 / Table 2 Sickness signs and corresponding alerts

病牛表现 / What you see in a sick animal	胶囊测到的信号 / What the bolus measures	平台告警 / Platform alert
发烧、体温升高 / Fever, elevated temperature	瘤胃温度连续升高、昼夜节律打乱 / Sustained rumen temperature rise, disturbed day-night rhythm	温度异常告警 / Temperature-abnormal alert
吃不下饭、反刍减少 / Loss of appetite, less rumination	瘤胃运动指数下降 / Drop in rumen motility index	消化异常告警 / Digestive-abnormal alert
饮水次数和时段打乱 / Drinking frequency and timing disrupted	饮水事件频次与分布变化 / Changes in drinking-event frequency and distribution	饮水异常提示 / Drinking-pattern alert
感染严重时综合恶化 / Overall deterioration when infection worsens	多项指标同时异常 / Several indicators off at once	人工智能综合异常分 / AI composite anomaly score

四类告警通过手机 App 和网页推送，直接指向具体的那头牛。牛仔拿到提醒后过去查伤口，确诊后用标准方法处理（清除幼虫、使用杀蛆剂，或注射伊维菌素类药物）。每发现一头、治好一头，伤口就少一个产卵点，疫情扩散就慢一分。 All four alert types arrive on the mobile app and web dashboard, pointing to the specific animal. Your crew inspects the wound, confirms, and applies standard treatment (larvae removal, approved larvicides or ivermectin-class injectables). Every animal found and treated early removes one egg-laying site and slows the outbreak.

2.2 对墨西哥牧场的实际价值 2.2 What this means on a Mexican ranch

- 大范围放牧、看不过来：典型牧场几天才巡一圈牛，正常情况下等发现时伤口已经很大、牛已消瘦；胶囊把发现时点提前到牛刚开始发热的时候。 / In extensive grazing where animals are checked every few days, detection normally

happens only once wounds are large and the animal is visibly losing condition. The bolus moves detection forward to when the fever starts.

- 疫情追踪：平台内置疫情工作台，可记录哪头牛在哪天确诊、伤口位置、严重程度、治疗进展，帮您向官方和买家展示牧场的防控工作。 / A built-in epidemic workbench logs each confirmed case, wound site, severity and treatment, demonstrating your control program to authorities and buyers.
- 同一套系统顺带管其他：体外寄生虫之外，这套监测同样覆盖其他发热性疾病（如边虫病、呼吸道疾病）、发情监测、围栏越界与定位。 / The same system also covers other febrile diseases (e.g., anaplasmosis, respiratory disease), estrus detection, fence breach and positioning.

三、我们不能做到的：先说清楚的五件事 3. What We Cannot Do: Five Honest Limitations

以下每一条都是我们主动说明的，请您在评估时把它们算进去。 We state these proactively; please factor them into your evaluation.

3.1 胶囊看不到伤口，也分不出是哪种病 3.1 No wound visibility, no disease-specific diagnosis

胶囊在瘤胃里，物理上看不到体表伤口和幼虫；它只能告诉您“这头牛不对劲”，而不能告诉您“这是蛆病”。发热可能来自任何疾病，最终确诊需要人工检查伤口。 Inside the rumen, the bolus physically cannot see skin wounds or larvae; it tells you “this animal is off”, never “this is screwworm”. Fever can stem from any disease — final confirmation requires wound inspection.

3.2 最早阶段检测不到 3.2 The earliest stage is invisible to any device

苍蝇刚产完卵的头一两天，牛体没有任何可测反应。胶囊的价值不是比兽医看得早，而是比“没人看”早：在没有人手每天逐头检查的牧场，它代替了那双不存在的眼睛。 In the first day or two after egg hatch there is nothing to measure. The value of the bolus is not seeing earlier than a veterinarian — it is seeing earlier than no one: on ranches without daily individual checks, it substitutes for eyes that are not there.

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3.3 幼牛无法使用胶囊 3.3 Calves are not covered by the bolus

新生犊牛的脐部伤口是蛆病最常见的侵袭部位之一，但幼牛瘤胃尚未发育完全，不能投放胶囊。针对幼牛，建议配合我们的耳标监测仪（测活动与体征）并加强产房卫生与新生犊牛伤口处理；请把这一点算入方案设计。 Navel wounds of newborn calves are among the most common invasion sites, yet calves cannot carry a bolus because their rumen is not fully developed. For calves we recommend our ear-tag monitor plus standard navel treatment and calving-area hygiene — please plan for this in your program design.

3.4 高温天气会干扰体温判断 3.4 Heat stress interferes with temperature judgment

墨西哥大部分地区气温高，热应激本身就会抬高牛的体温基线，容易和发烧混淆。我们的系统会按牛的个体基线而非固定阈值判断异常，但在极端高温季节，误报率会上升，需要在试点阶段用当地数据调参。这是我们选择“先试点、后推广”的原因之一。 In most of Mexico, heat stress itself raises the temperature baseline and can be confused with fever. Our system judges against each animal's own baseline rather than a fixed threshold, but during extreme heat the false-alarm rate rises until parameters are tuned with local data. That is why we recommend pilot first, then scale.

3.5 效果数字需要试点验证，不预先承诺 3.5 Performance numbers come from the pilot, not promises

我们不在这份材料里给出“检出率 99%”之类的承诺。原因很简单：这套系统尚未在蛆病疫情中做过正式验证，任何承诺都是不负责任的。我们建议通过试点拿到真实数字：在疫区牧场选取感染牛与健康对照牛各若干头，记录每次人工发现的日期，再回看系统是否更早发出了提醒、提前了多少小时、误报有多少——拿到数字之后再谈大规模推广。 We will not promise figures like “99% detection” here. The reason is simple: this system has not yet been formally validated in a screwworm outbreak, and any such promise would be irresponsible. We propose getting real numbers through a pilot: enroll infected and healthy control animals in an affected ranch, log every manual discovery date, then check whether the system alerted earlier, by how many hours, and at what false-alarm rate — and only then discuss scale-up.

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四、建议的下一步：一个试点牧场 4. Recommended Next Step: One Pilot Ranch

表 3 试点方案概要 / Table 3 Pilot outline

项目 / Item	安排 / Arrangement
场地 / Site	墨西哥疫区的一座商业牧场 / One commercial ranch in an affected region of Mexico
动物 / Animals	成年牛佩戴胶囊（感染组 + 健康对照组）；幼牛用耳标监测仪 / Adult cattle wear boluses (infected group + healthy controls); calves wear ear tags
记录 / Records	人工发现每头病牛的日期、伤口位置、严重度、治疗日 / Manual discovery date, wound site, severity and treatment date for every case
评估 / Evaluation	系统提醒是否早于人工发现、提前多少小时、误报率多少 / Whether alerts preceded manual checks, by how many hours, at what false-alarm rate
结论 / Outcome	拿到真实效果数字，再评估大规模推广与出口档案应用 / Real performance numbers, then assess scale-up and export-dossier use

试点期间我们提供设备、安装指导、培训和远程支持；具体的设备数量、期限与费用安排，欢迎另行商谈。 During the pilot we provide devices, installation guidance, training and remote support; commercial terms (quantities, duration, costs) are open for discussion.

五、结语 5. Closing

蛆病之所以难防，是因为它藏在伤口里、长在大地上；而我们能做的，是把每头牛的体温和行为变成每天可查的记录，让牛仔的眼睛长在每一头牛身上。它不是魔法，但在一位牛仔要看五百头牛的牧场里，它是真实可用的帮手。欢迎您们就试点方案与我们进一步沟通。 Screwworm is hard to fight because it hides in wounds and roams across the land. What we can do is turn every animal's temperature and behaviour into a daily record — putting eyes on every head of cattle. It is not magic, but on a ranch where one cowboy watches five hundred animals, it is a genuinely useful pair of hands. We welcome your questions about the pilot.

此致，并期待合作。 We remain at your disposal and look forward to cooperating.

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